

Lab #3. Django Book Store with Postgres

YKumar, 2024-6-15

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0. Intro to Django

Django is a web framework; it is a collection of tools and libraries packaged together to help developers build web applications more efficiently.

Key Features:

Object-relational mapping (ORM): a technique that allows developers to interact **with a database** using an object-oriented paradigm **instead of writing raw SQL queries**.

URL routing: allows developers to define **routes** for different parts of the app, ensuring that specific URLs trigger the correct functions or methods.

HTML templating: allows the dynamic generation of HTML content by combining static HTML with app data.

Form handling provides tools to handle web forms, including form validation and processing, simplifies the process of capturing **user input through forms**, validating the input, and storing or using the input within the application.

Unit testing: helps in maintaining code quality and reliability by allowing developers to test their code in isolation and catch bugs early in the development process.

Observe <https://www.djangoproject.com/> for more information

1. Set up

1.1. Install latest version of Python

- at **python.org** -> Downloads -> Download -> Find in our Downloads and Run, allow to upgrade
- Create a folder called **django**, select it in VSC, Terminal -> New Terminal, type

```
\3320\django> pip install django
```

If you get any errors make sure to run this with administrator privileges: you can add **sudo** in front of your command on mac, run on PowerShell (run as administrator) if you are on Windows. To get to powershell type power in the search->right-click->Run as administrator

Windows: **pip install --upgrade --force-reinstall django**

Mac: **sudo pip install --upgrade --force-reinstall django**

```
Administrator: Windows PowerShell
PS C:\Users\galae\Desktop> cd '00472-337513824 GS Backup 12.20.22 JC'
PS C:\Users\galae\Desktop\00472-337513824 GS Backup 12.20.22 JC> cd Kean_Lectures
PS C:\Users\galae\Desktop\00472-337513824 GS Backup 12.20.22 JC\Kean_Lectures> cd 3320
PS C:\Users\galae\Desktop\00472-337513824 GS Backup 12.20.22 JC\Kean_Lectures\3320> cd django
PS C:\Users\galae\Desktop\00472-337513824 GS Backup 12.20.22 JC\Kean_Lectures\3320\django> pip install --upgrade --force-reinstall django
Collecting django
  Using cached Django-5.0.6-py3-none-any.whl.metadata (4.1 kB)
Collecting asgiref<4,>=3.7.0 (from django)
  Using cached asgiref-3.8.1-py3-none-any.whl.metadata (9.3 kB)
Collecting sqlparse>=0.3.1 (from django)
  Using cached sqlparse-0.5.0-py3-none-any.whl.metadata (3.9 kB)
Collecting tzdata (from django)
  Using cached tzdata-2024.1-py2.py3-none-any.whl.metadata (1.4 kB)
Using cached Django-5.0.6-py3-none-any.whl (8.2 MB)
Using cached asgiref-3.8.1-py3-none-any.whl (23 kB)
Using cached sqlparse-0.5.0-py3-none-any.whl (43 kB)
Using cached tzdata-2024.1-py2.py3-none-any.whl (345 kB)
Installing collected packages: tzdata, sqlparse, asgiref, django
  Attempting uninstall: tzdata
    Found existing installation: tzdata 2024.1
    Uninstalling tzdata-2024.1:
      Successfully uninstalled tzdata-2024.1
  Attempting uninstall: sqlparse
    Found existing installation: sqlparse 0.5.0
    Uninstalling sqlparse-0.5.0:
      Successfully uninstalled sqlparse-0.5.0
Successfully installed asgiref-3.8.1 django-5.0.6 sqlparse-0.5.0 tzdata-2024.1
PS C:\Users\galae\Desktop\00472-337513824 GS Backup 12.20.22 JC\Kean_Lectures\3320\django>
```

1.2. Install psycopg2

This is the PostgreSQL adapter for Python:

pip install psycopg2-binary

- PS C:\Users\galae\Desktop\00472-337513824 GS Backup 12.20.22 JC\KEAN_Lectures\3320\django> pip install psycopg2-binary
Collecting psycopg2-binary
 Downloading psycopg2_binary-2.9.9-cp312-cp312-win_amd64.whl.metadata (4.6 kB)
 Downloading psycopg2_binary-2.9.9-cp312-cp312-win_amd64.whl (1.2 MB)
 1.2/1.2 MB 8.2 MB/s eta 0:00:00
Installing collected packages: psycopg2-binary
Successfully installed psycopg2-binary-2.9.9

1.3. Create a new project

django-admin startproject bookstore

cd bookstore

- PS C:\Users\galae\Desktop\00472-337513824 GS Backup 12.20.22 JC\KEAN_Lectures\3320\django> django-admin startproject bookstore
 - PS C:\Users\galae\Desktop\00472-337513824 GS Backup 12.20.22 JC\KEAN_Lectures\3320\django> cd bookstore
 - PS C:\Users\galae\Desktop\00472-337513824 GS Backup 12.20.22 JC\KEAN_Lectures\3320\django\bookstore> ls
- Directory: C:\Users\galae\Desktop\00472-337513824 GS Backup 12.20.22 JC\KEAN_Lectures\3320\django\bookstore
- | Mode | LastWriteTime | Length | Name |
|--------|--------------------|--------|-----------|
| d---- | 6/15/2024 11:33 AM | | bookstore |
| -a---- | 6/15/2024 11:33 AM | 687 | manage.py |

Task 1. Provide a screenshot of your result.

2. PostgreSQL Database set up

2.1. Download

Open the link: <https://www.enterprisedb.com/downloads/postgres-postgresql-downloads>

Download PostgreSQL

Open source PostgreSQL packages and installers from EDB

PostgreSQL Version	Linux x86-64	Linux x86-32	Mac OS X	Windows x86-64	Windows x86-32
16.3	postgresql.org	postgresql.org			Not supported

Your download should begin in a few seconds

Click here if your download does not start automatically.

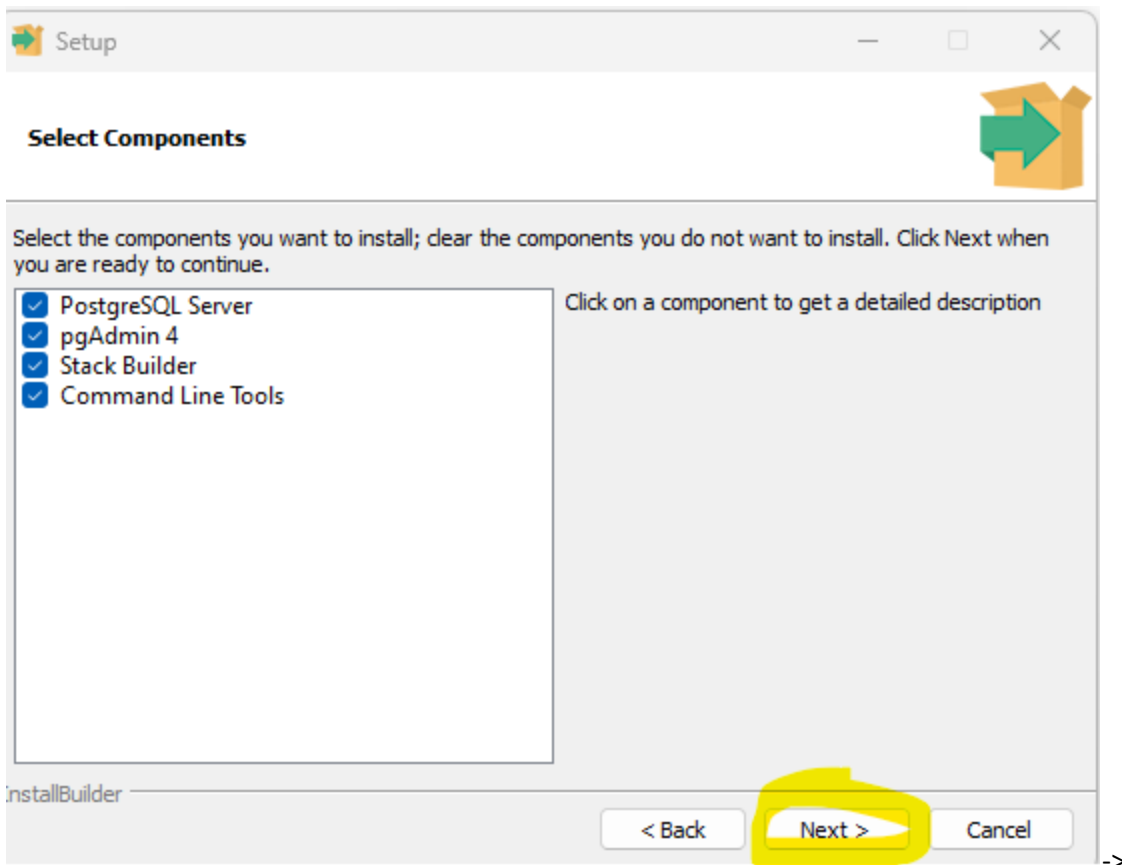
Wait or click

▼ Today

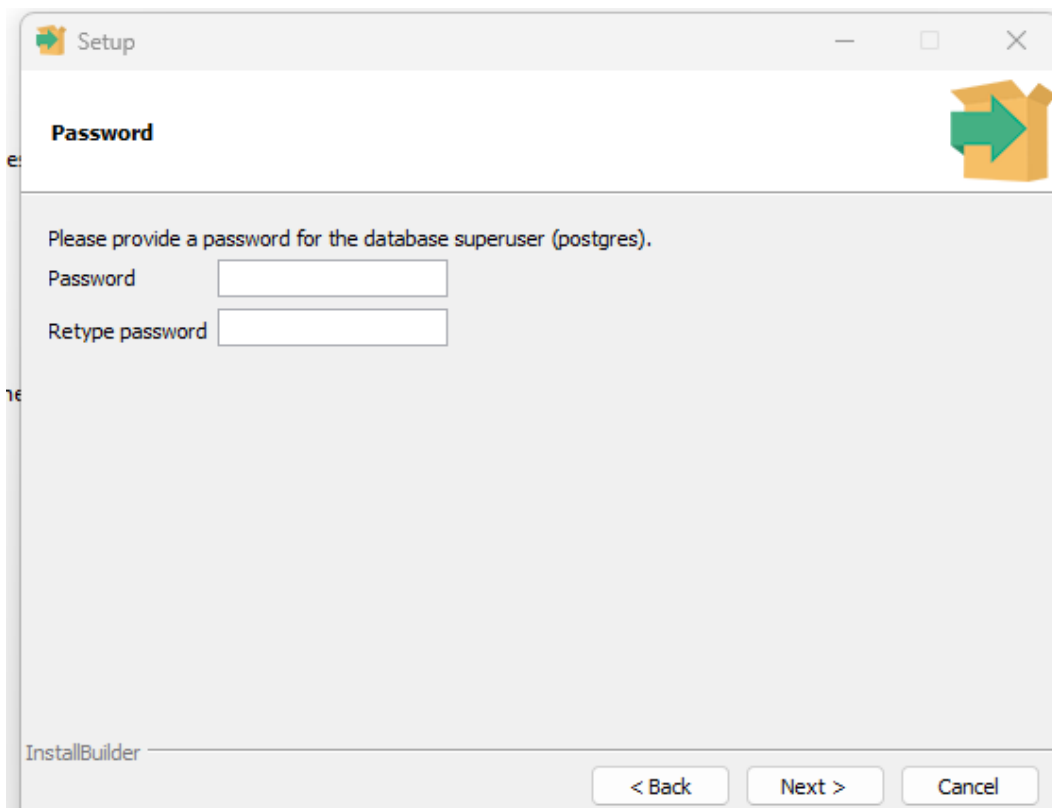
 postgresql-16.3-2-windows-x64

2.2. Install

Run the .exe file. Proceed with Yes-> Next->Next->

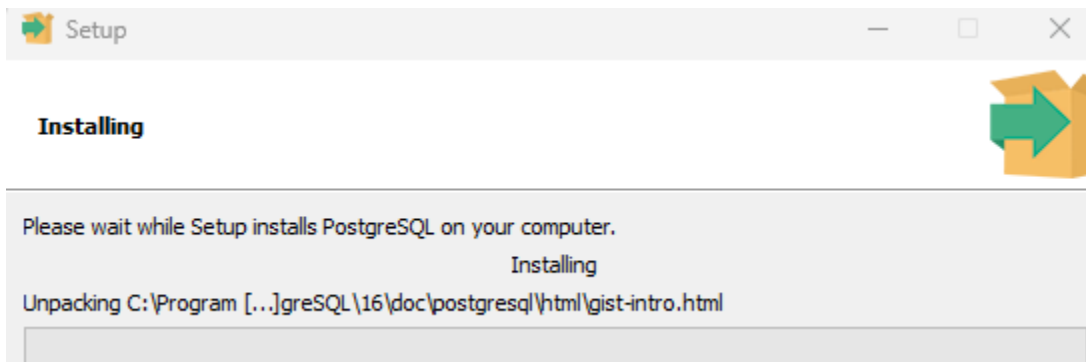


Create a password, remember what you put



Mine: **CP\$3320\$ummer!**

Keep the port, locale and other settings default (just click next)



For mac only

Use homebrew for Mac: <https://brew.sh/>

Run `/bin/bash -c "$(curl -fsSL`

`https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"`

`brew install postgresql`

`brew services start postgresql`

Run `psql postgres`

3. Download pgadmin

3.1. download and unpack

Download from <https://www.pgadmin.org/download/>

pgAdmin 4 (Windows)

Download

Maintainer: pgAdmin Development Team

pgAdmin is available for 64 bit Windows™ 7 SP1 (desktop) or 2008R2 (server) and above, up to v4.30.

v5.0 and later are supported on Windows 8 (desktop) or 2012 (server) and above.

v7.0 and later are supported on Windows 10 (desktop) or 2016 (server) and above.

32 bit Windows support is available for versions up to v4.29.

The packages below include both the Desktop Runtime and Web Application:

- 📄 pgAdmin 4 v8.8 (released June 6, 2024)
- 📄 pgAdmin 4 v8.7 (released May 30, 2024)
- 📄 pgAdmin 4 v8.6 (released May 2, 2024)
- 📄 pgAdmin 4 v7.8 (released Oct. 19, 2023)
- 📄 pgAdmin 4 v6.21 (released March 9, 2023)

📄 **CURRENT_MAINTAINER**
📄 **pgadmin4-8.8-x64.exe**

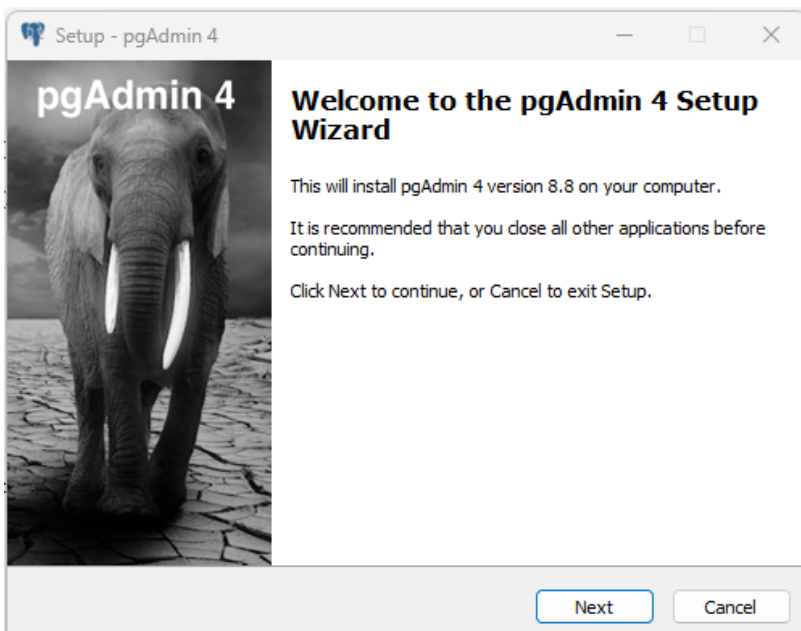
June 6, 2024, 9:30 a.m.

136 bytes

June 6, 2024, 9:31 a.m.

183.0 MB

📄 pgadmin4-8.8-x64



Accept license and proceed with defaults

Find and open pgadmin on your machine - > run as administrator

I used the same fake password

Migrate Saved Passwords

Please enter your master password.

pgAdmin now stores any saved passwords in Windows WinVaultKeyring. Enter the master password for your existing pgAdmin saved passwords and they will be migrated to the operating system store when you click OK.

 ?

Reset Master Password

✓ OK

I decided to reset this

Welcome



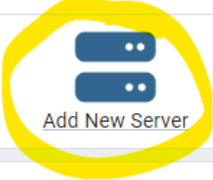
pgAdmin

Management Tools for PostgreSQL

Feature rich | Maximises PostgreSQL | Open Source

pgAdmin is an Open Source administration and management tool for the PostgreSQL database. It includes a graphical administration interface, an SQL query tool, a procedural code debugger and much more. The tool is designed to answer the needs of developers, DBAs and system administrators alike.

Quick Links



Add New Server



[Configure pgAdmin](#)

Getting Started



[PostgreSQL Documentation](#)



[pgAdmin Website](#)



[Planet PostgreSQL](#)



[Community Support](#)

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents × Processes >

Register - Server ×

General

Connection

Parameters

SSH Tunnel

Advanced

Name

YKBook

Server group

Servers | ▾

Background

×

Foreground

×

Connect now?

☒

Comments

for 3320

ⓘ ⓘ

Close

Reset

Save

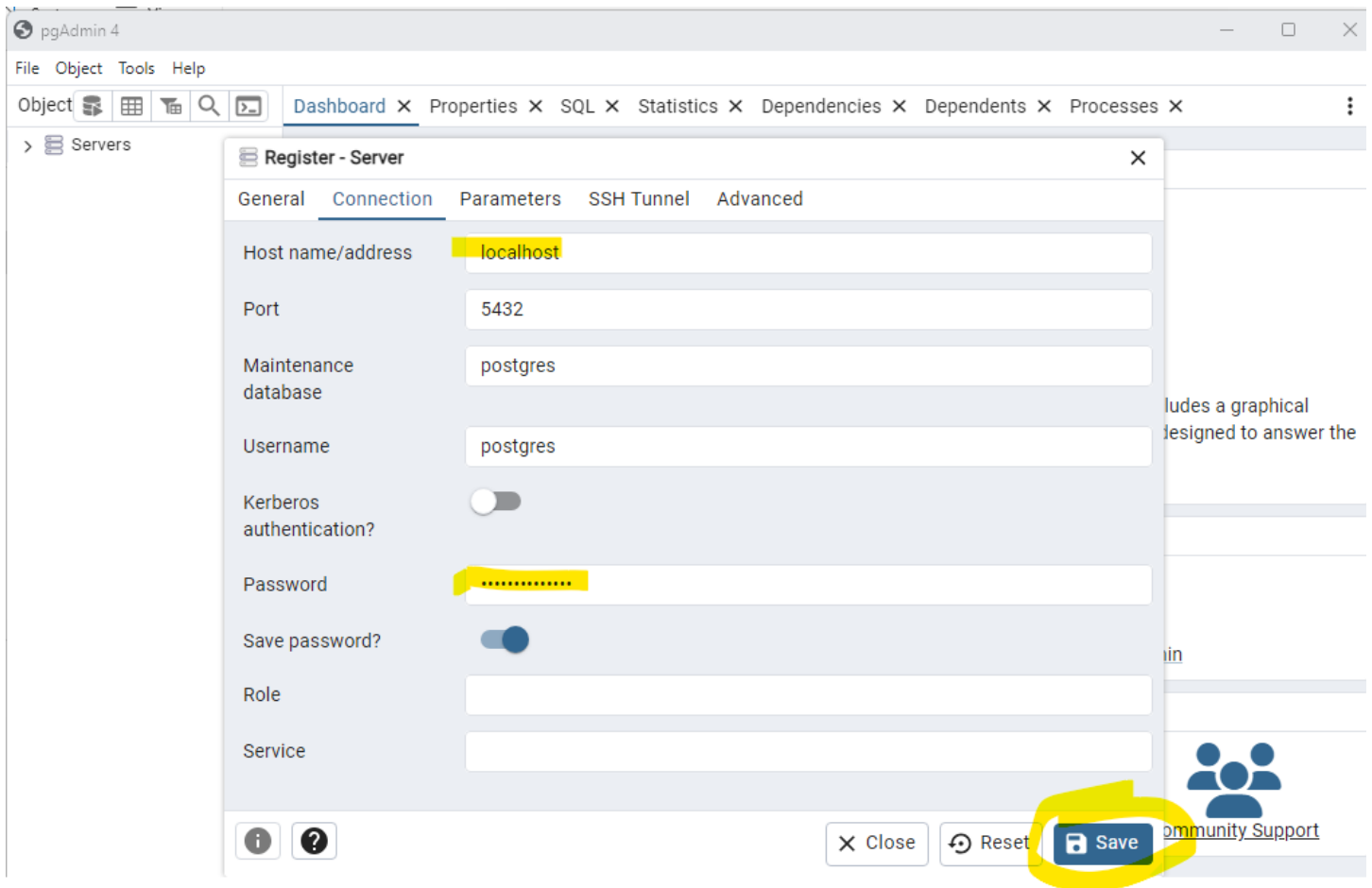
ⓘ

ⓘ

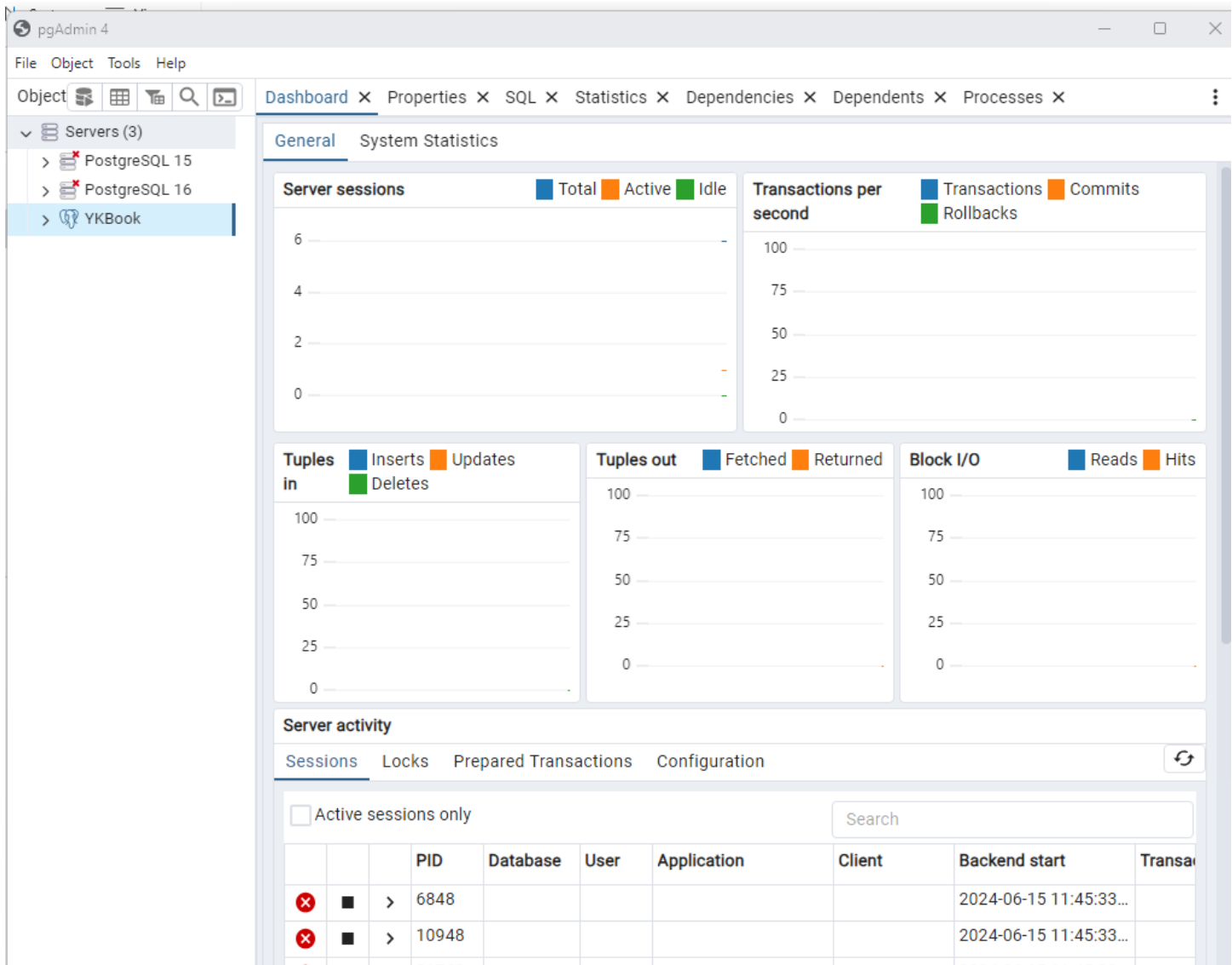
×

 Either Host name or Service must be specified.

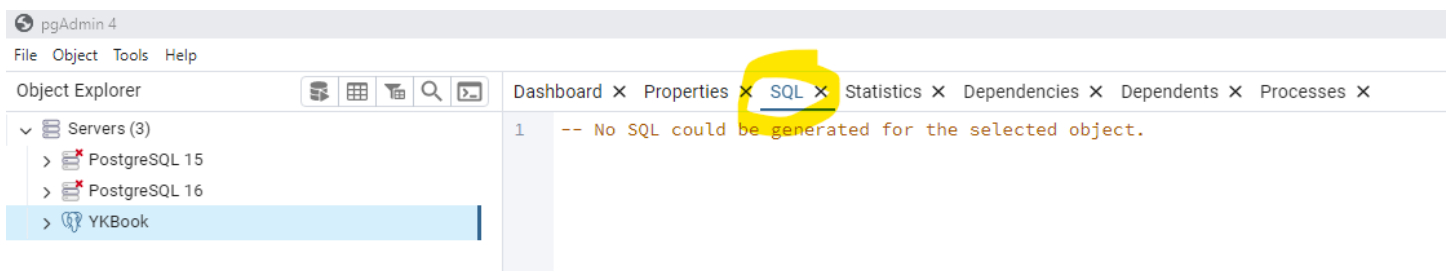
×



The result



Go to SQL



Task 2. Provide screenshots of your download results.

4. SQL queries

4.1. Create Database

Use your initials to create this database

CREATE DATABASE YKBookStore (use your initial)

Create - Database

×

GeneralDefinitionSecurityParametersAdvancedSQL

Database

YKBookStore

OID

Owner

postgres

▼

Comment

CPS3320

ⓘ

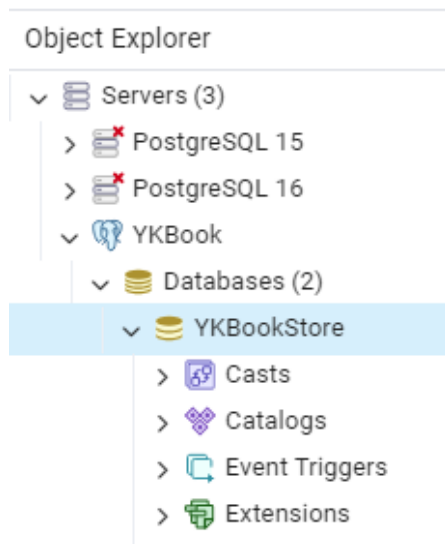
?

✕ Close

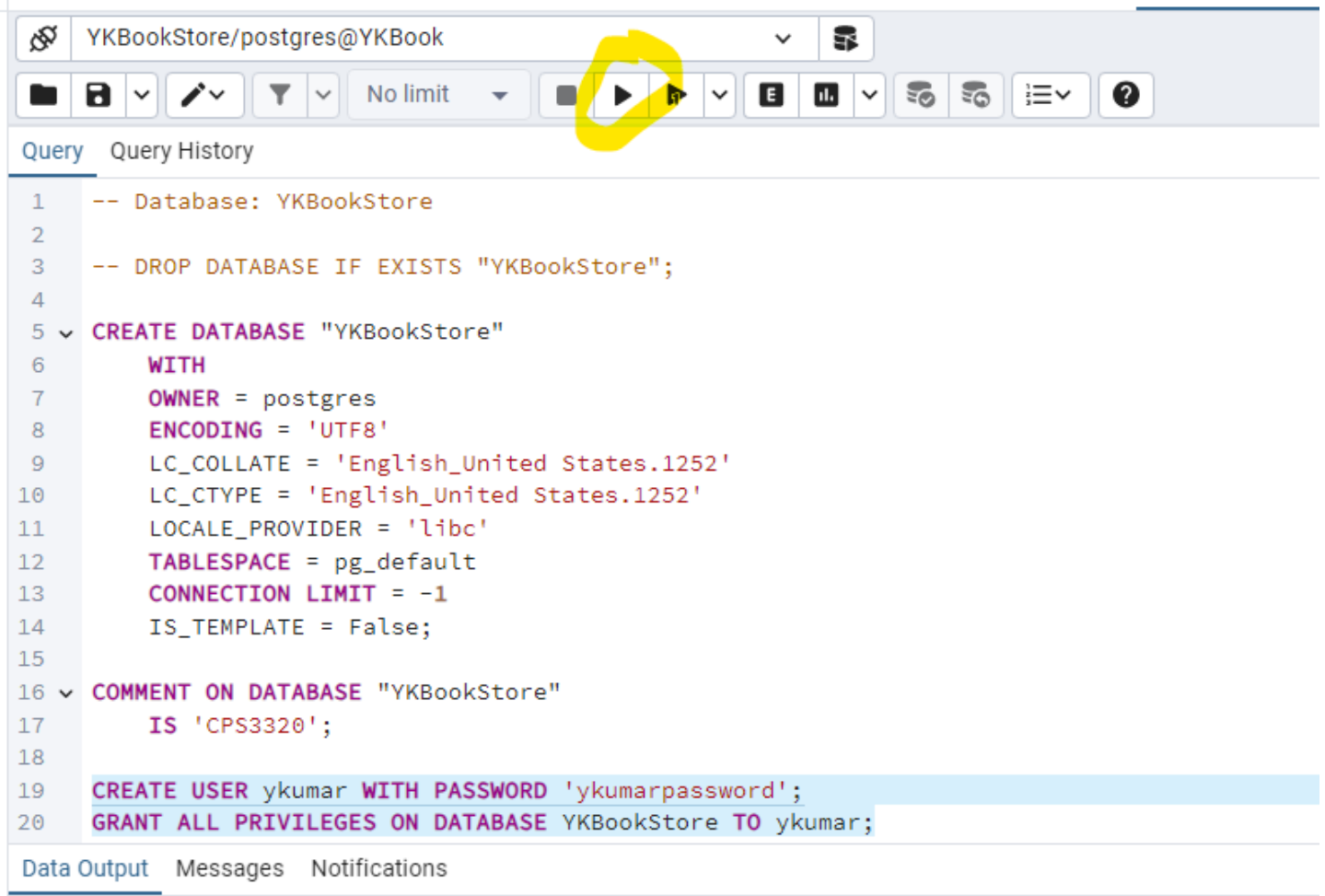
↺ Reset

💾 Save

Right-click on Database -> create script



Create a user



YKBookStore/postgres@YKBook

Query Query History

```
1  -- Database: YKBookStore
2
3  -- DROP DATABASE IF EXISTS "YKBookStore";
4
5  CREATE DATABASE "YKBookStore"
6      WITH
7      OWNER = postgres
8      ENCODING = 'UTF8'
9      LC_COLLATE = 'English_United States.1252'
10     LC_CTYPE = 'English_United States.1252'
11     LOCALE_PROVIDER = 'libc'
12     TABLESPACE = pg_default
13     CONNECTION LIMIT = -1
14     IS_TEMPLATE = False;
15
16  COMMENT ON DATABASE "YKBookStore"
17      IS 'CPS3320';
18
19  CREATE USER ykumar WITH PASSWORD 'ykumarpasssword';
20  GRANT ALL PRIVILEGES ON DATABASE YKBookStore TO ykumar;
```

Data Output Messages Notifications

4.2: Create Tables

Right-click on Tables – choose Query Tool

The screenshot shows a database management interface with a left sidebar displaying the database structure. The 'YKBookStore' database is selected, showing various objects like Casts, Catalogs, Event Triggers, Extensions, Foreign Data Wrappers, Languages, Publications, and Schemas. The 'public' schema is expanded, showing objects like Aggregates, Collations, Domains, FTS Configurations, FTS Dictionaries, FTS Parsers, FTS Templates, Foreign Tables, Functions, Materialized Views, Operators, Procedures, Sequences, and Tables. The 'Tables' object is highlighted.

The main pane shows the SQL query editor with the following commands:

```

1 CREATE TABLE authors (
2     author_id SERIAL PRIMARY KEY,
3     name VARCHAR(100) NOT NULL,
4     birthdate DATE,
5     nationality VARCHAR(50)
6 );
7
8 CREATE TABLE books (
9     book_id SERIAL PRIMARY KEY,
10    title VARCHAR(200) NOT NULL,
11    author_id INTEGER REFERENCES authors(author_id),
12    genre VARCHAR(50),
13    price DECIMAL(10, 2),
14    publisher VARCHAR(100)
15 );
16
17 CREATE TABLE videos (
18     video_id SERIAL PRIMARY KEY,
19     title VARCHAR(200) NOT NULL,
20     genre VARCHAR(50),
21     price DECIMAL(10, 2),
22     publisher VARCHAR(100)
23 );
24

```

The 'Data Output' tab is selected, showing the message: 'Query returned successfully in 73 msec.'

```

CREATE TABLE authors (
    author_id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    birthdate DATE,
    nationality VARCHAR(50)
);

```

```

CREATE TABLE books (
    book_id SERIAL PRIMARY KEY,
    title VARCHAR(200) NOT NULL,
    author_id INTEGER REFERENCES authors(author_id),
    genre VARCHAR(50),
    price DECIMAL(10, 2),
    publisher VARCHAR(100)
);

```

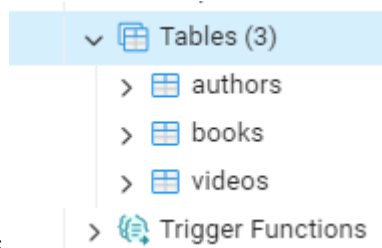
```

CREATE TABLE videos (
    video_id SERIAL PRIMARY KEY,
    title VARCHAR(200) NOT NULL,
    genre VARCHAR(50),

```

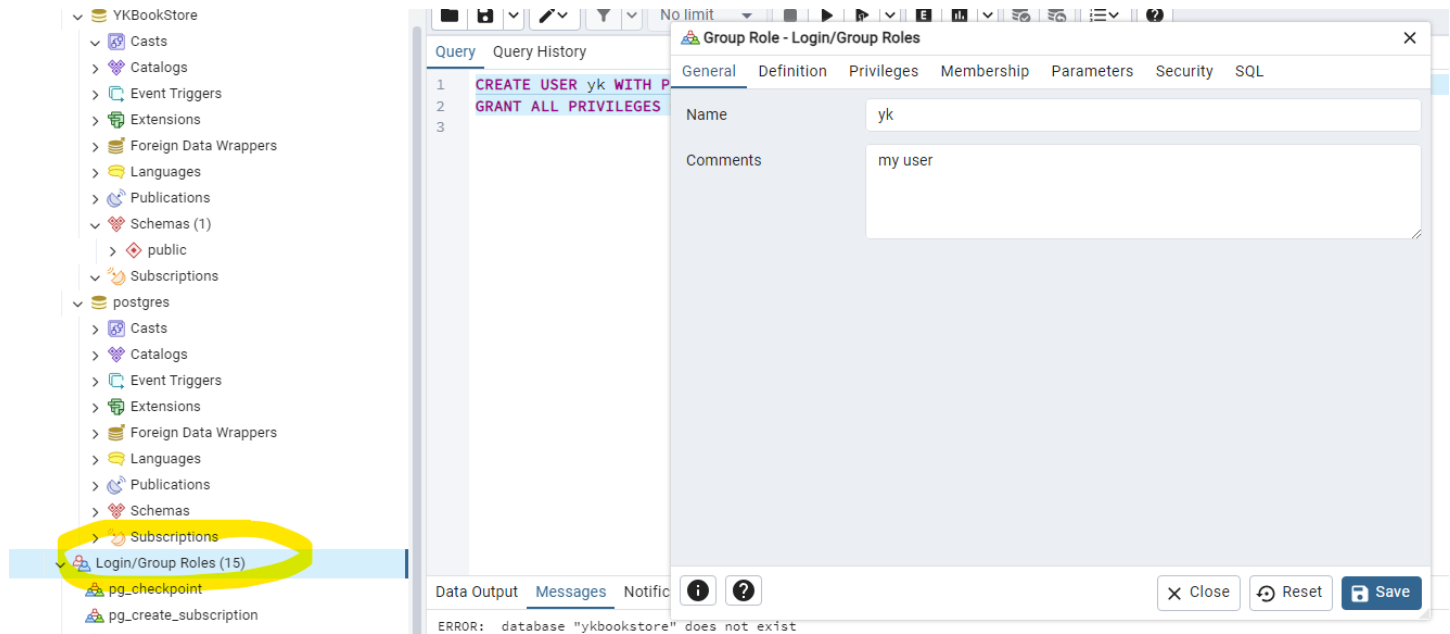
price DECIMAL(10, 2),
publisher VARCHAR(100)

);



Observe the results

4.3: Create a User



Group Role - yk

General Definition Privileges Membership

Password

Account expires
Please note that if you

Connection limit

Group Role - yk

General Definition Privileges Membership

Can login? ☒

Superuser? ☒

Create roles? ☒

Create databases? ☒

Inherit rights from the parent roles? ☒

Can initiate streaming replication and backups? ☐

Bypass RLS? ☒

Save every time

4.4: Populate Tables

4.4.1. adjustments to the tables

Add the missing isbn column to the books table:

ALTER TABLE books

ADD COLUMN isbn VARCHAR(20);

Add the missing release_date column to the videos table:

ALTER TABLE videos

ADD COLUMN release_date DATE;

4.4.2. Populate tables

Adjust as much as you like

INSERT INTO authors (author_id, name, birthdate, nationality) VALUES

(1, 'James Patterson', '1947-03-22', 'American'),

(2, 'J.K. Rowling', '1965-07-31', 'British'),

(3, 'Stephen King', '1947-09-21', 'American'),

(4, 'John Grisham', '1955-02-08', 'American'),

(5, 'Agatha Christie', '1890-09-15', 'British'),
 (6, 'George R.R. Martin', '1948-09-20', 'American'),
 (7, 'Dan Brown', '1964-06-22', 'American'),
 (8, 'Jane Austen', '1775-12-16', 'British'),
 (9, 'Mark Twain', '1835-11-30', 'American'),
 (10, 'Charles Dickens', '1812-02-07', 'British');

INSERT INTO books (book_id, title, author_id, genre, price, publisher, isbn) VALUES
 (1, 'Along Came a Spider', 1, 'Thriller', 12.99, 'Little, Brown', '9780316693646'),
 (2, 'Harry Potter and the Sorcerer's Stone', 2, 'Fantasy', 9.99, 'Scholastic',
 '9780439708180'),
 (3, 'The Shining', 3, 'Horror', 14.99, 'Doubleday', '9780385121675'),
 (4, 'The Firm', 4, 'Legal Thriller', 10.99, 'Doubleday', '9780385416344'),
 (5, 'Murder on the Orient Express', 5, 'Mystery', 7.99, 'Collins Crime Club', '9780007119318'),
 (6, 'A Game of Thrones', 6, 'Fantasy', 15.99, 'Bantam Spectra', '9780553103540'),
 (7, 'The Da Vinci Code', 7, 'Mystery', 12.99, 'Doubleday', '9780385504201'),
 (8, 'Pride and Prejudice', 8, 'Romance', 5.99, 'T. Egerton', '9781503290563'),
 (9, 'The Adventures of Tom Sawyer', 9, 'Adventure', 6.99, 'American Publishing Company',
 '9780486400778'),
 (10, 'A Tale of Two Cities', 10, 'Historical Fiction', 8.99, 'Chapman & Hall', '9781853260391');

INSERT INTO videos (video_id, title, genre, price, publisher, release_date) VALUES
 (1, 'Inception', 'Sci-Fi', 14.99, 'Warner Bros.', '2010-07-16'),
 (2, 'The Shawshank Redemption', 'Drama', 9.99, 'Columbia Pictures', '1994-09-22'),
 (3, 'The Godfather', 'Crime', 12.99, 'Paramount Pictures', '1972-03-24'),
 (4, 'The Dark Knight', 'Action', 13.99, 'Warner Bros.', '2008-07-18'),
 (5, 'Pulp Fiction', 'Crime', 11.99, 'Miramax', '1994-10-14'),
 (6, 'Forrest Gump', 'Drama', 10.99, 'Paramount Pictures', '1994-07-06'),
 (7, 'Fight Club', 'Drama', 9.99, '20th Century Fox', '1999-10-15'),
 (8, 'The Matrix', 'Sci-Fi', 12.99, 'Warner Bros.', '1999-03-31'),
 (9, 'The Lord of the Rings: The Fellowship of the Ring', 'Fantasy', 15.99, 'New Line Cinema',
 '2001-12-19'),
 (10, 'Star Wars: Episode IV - A New Hope', 'Sci-Fi', 14.99, '20th Century Fox', '1977-05-25');

4.4.3. Verify by highlighting one at a time

Query

Query History

1

SELECT * FROM authors;

2

SELECT * FROM books;

3

SELECT * FROM videos;

4

Data Output

Messages

Notifications

≡+

▼

▼

SQL

	video_id [PK] integer	title character varying (200)	genre character varying (50)	price numeric (10,2)	publisher character varying (100)	release_date date
1	1	Inception	Sci-Fi	14.99	Warner Bros.	2010-07-16
2	2	The Shawshank Redemption	Drama	9.99	Columbia Pictures	1994-09-22
3	3	The Godfather	Crime	12.99	Paramount Pictures	1972-03-24
4	4	The Dark Knight	Action	13.99	Warner Bros.	2008-07-18
5	5	Pulp Fiction	Crime	11.99	Miramax	1994-10-14
6	6	Forrest Gump	Drama	10.99	Paramount Pictures	1994-07-06
7	7	Fight Club	Drama	9.99	20th Century Fox	1999-10-15
8	8	The Matrix	Sci-Fi	12.99	Warner Bros.	1999-03-31
9	9	The Lord of the Rings: The Fellowship of the Ring	Fantasy	15.99	New Line Cinema	2001-12-19
10	10	Star Wars: Episode IV - A New Hope	Sci-Fi	14.99	20th Century Fox	1977-05-25

Task 3. Adjust the list and add at least 1 author and least 1 book and at least 1 movie that were not there before. Take screenshots.

5. Connect to the database

5.1. Set up an environmental variable

In search on your computer Find and Click Environmental Variables **Environment Variables => System Variables**
→ **edit** Find postgres folder on your machine, save a path to bin: **C:\Program Files\PostgreSQL\16\bin**

> This PC > Windows (C:) > Program Files > PostgreSQL > 16 > bin

5.2. Run commands in VSC

Close and reopen VSC

psql -U postgres

in VSC

Use the password you created during installation

```
postgres
PS C:\Users\galae\Desktop\00472-337513824 GS Backup 12.20.22 JC\KEAN_Lectures\3320\django> psql -U postgres
Password for user postgres:
psql (16.3)
WARNING: Console code page (437) differs from Windows code page (1252)
8-bit characters might not work correctly. See psql reference
page "Notes for Windows users" for details.
Type "help" for help.

postgres=#
```

5.3. View Databases

```
postgres-# \l
```

List of databases								
Name	Owner	Encoding	Locale Provider	Collate	Ctype	ICU Locale	ICU Rules	Access privileges
YKBookStore	postgres	UTF8	libc	English_United States.1252	English_United States.1252			
postgres	postgres	UTF8	libc	English_United States.1252	English_United States.1252			
template0	postgres	UTF8	libc	English_United States.1252	English_United States.1252			=c
/postgres	+							po
stgres=CTc/postgres	postgres	UTF8	libc	English_United States.1252	English_United States.1252			=c
/postgres	+							po

(4 rows)

```
postgres-#
```

5.4. View Tables

```
postgres-# \c YKBookStore
You are now connected to database "YKBookStore" as user "postgres".
YKBookStore-# \dt
```

List of relations			
Schema	Name	Type	Owner
public	authors	table	postgres
public	books	table	postgres
public	videos	table	postgres

(3 rows)

```
YKBookStore-#
```

5.5. View Books

```
YKBookStore=# SELECT * FROM books;
```

book_id	title	author_id	genre	price	publisher	isbn
1	Along Came a Spider	1	Thriller	12.99	Little, Brown	9780316693646
2	Harry Potter and the Sorcerers Stone	2	Fantasy	9.99	Scholastic	9780439708180
3	The Shining	3	Horror	14.99	Doubleday	9780385121675
4	The Firm	4	Legal Thriller	10.99	Doubleday	9780385416344
5	Murder on the Orient Express	5	Mystery	7.99	Collins Crime Club	9780007119318
6	A Game of Thrones	6	Fantasy	15.99	Bantam Spectra	9780553103540
7	The Da Vinci Code	7	Mystery	12.99	Doubleday	9780385504201
8	Pride and Prejudice	8	Romance	5.99	T. Egerton	9781503290563
9	The Adventures of Tom Sawyer	9	Adventure	6.99	American Publishing Company	9780486400778
10	A Tale of Two Cities	10	Historical Fiction	8.99	Chapman & Hall	9781853260391

(10 rows)

```
YKBookStore=#
```

5.6. Create more tables

```
CREATE TABLE students (student_id SERIAL PRIMARY KEY, name VARCHAR(255), email VARCHAR(255) UNIQUE, birthdate DATE);
```

```
CREATE TABLE transactions (transaction_id SERIAL PRIMARY KEY, student_id INTEGER
REFERENCES students(student_id), book_id INTEGER REFERENCES books(book_id), video_id
INTEGER REFERENCES videos(video_id), borrowed_date DATE DEFAULT CURRENT_DATE, due_date
DATE);
```

```
YKBookStore=# CREATE TABLE students (student_id SERIAL PRIMARY KEY, name VARCHAR(255), email VARCHAR(255) UNIQUE, birthdate DATE);
ERROR: relation "students" already exists
YKBookStore=# CREATE TABLE transactions (transaction_id SERIAL PRIMARY KEY, student_id INTEGER REFERENCES students(student_id), book_id INTEGE
R REFERENCES books(book_id), video_id INTEGER REFERENCES videos(video_id), borrowed_date DATE DEFAULT CURRENT_DATE, due_date DATE);
CREATE TABLE
YKBookStore=#
```

5.7. Populate new tables

Make sure to add your name to this list

```
INSERT INTO students (name, email, birthdate) VALUES
('Yulia Kumar', 'ykumar@kean.edu', '2024-06-15'),
('Jane Smith', 'jane.smith@example.com', '1997-08-15'),
('Alice Johnson', 'alice.johnson@example.com', '1999-02-10'),
('Bob Brown', 'bob.brown@example.com', '2000-12-01'),
('Eve Davis', 'eve.davis@example.com', '1996-07-22');
```

Adjust as you like

```
INSERT INTO transactions (student_id, book_id, video_id, borrowed_date, due_date) VALUES
(1, 1, NULL, '2023-06-01', '2024-05-15'),
(2, NULL, 1, '2023-06-03', '2024-05-17'),
(3, 3, NULL, '2023-06-05', '2024-05-19'),
(4, NULL, 3, '2023-06-07', '2024-05-21'),
(5, 5, NULL, '2023-06-09', '2024-05-23'),
(1, NULL, 2, '2023-06-11', '2024-05-25'),
(2, 2, NULL, '2023-06-13', '2024-05-27'),
(3, NULL, 4, '2023-06-15', '2024-05-29'),
(4, 4, NULL, '2023-06-17', '2024-06-01'),
(5, NULL, 5, '2023-06-19', '2024-06-03');
```

```

YKBookStore=# INSERT INTO students (name, email, birthdate) VALUES
YKBookStore=# ('Yulia Kumar', 'ykumar@kean.edu', '2024-06-15'),
YKBookStore=# ('Jane Smith', 'jane.smith@example.com', '1997-08-15'),
YKBookStore=# ('Alice Johnson', 'alice.johnson@example.com', '1999-02-10'),
YKBookStore=# ('Bob Brown', 'bob.brown@example.com', '2000-12-01'),
YKBookStore=# ('Eve Davis', 'eve.davis@example.com', '1996-07-22');
INSERT 0 5
YKBookStore=#
YKBookStore=# INSERT INTO transactions (student_id, book_id, video_id, borrowed_date, due_date) VALUES
YKBookStore=# (1, 1, NULL, '2023-06-01', '2024-05-15'),
YKBookStore=# (2, NULL, 1, '2023-06-03', '2024-05-17'),
YKBookStore=# (3, 3, NULL, '2023-06-05', '2024-05-19'),
YKBookStore=# (4, NULL, 3, '2023-06-07', '2024-05-21'),
YKBookStore=# (5, 5, NULL, '2023-06-09', '2024-05-23'),
YKBookStore=# (1, NULL, 2, '2023-06-11', '2024-05-25'),
YKBookStore=# (2, 2, NULL, '2023-06-13', '2024-05-27'),
YKBookStore=# (3, NULL, 4, '2023-06-15', '2024-05-29'),
YKBookStore=# (4, 4, NULL, '2023-06-17', '2024-06-01'),
YKBookStore=# (5, NULL, 5, '2023-06-19', '2024-06-03');
INSERT 0 10
YKBookStore=# 

```

5.8. Verify

```

YKBookStore=# select * from students;

```

student_id	name	email	birthdate
1	Yulia Kumar	ykumar@kean.edu	2024-06-15
2	Jane Smith	jane.smith@example.com	1997-08-15
3	Alice Johnson	alice.johnson@example.com	1999-02-10
4	Bob Brown	bob.brown@example.com	2000-12-01
5	Eve Davis	eve.davis@example.com	1996-07-22

(5 rows)

```

YKBookStore=# 

```

Task 4. Provide a screenshot like 5.8. User(s) on top of the table should be you (your teammates) names

6. Set up the app

6.1. Update settings.py

Replace the existing DATABASES configuration with the following:

```

DATABASES = {
    "default": {
        "ENGINE": "django.db.backends.postgresql",
        "NAME": "YKBookStore",
        "USER": "postgres",

```

```

    "PASSWORD": "yourpassword", # replace with your actual password
    "HOST": "localhost",
    "PORT": "5432",
}
}

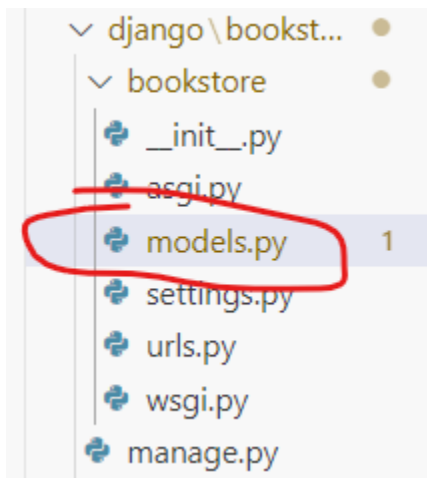
73 # Database
74 # https://docs.djangoproject.com/en/5.0/ref/settings/#databases
75 DATABASES = {
76     "default": {
77         "ENGINE": "django.db.backends.postgresql",
78         "NAME": "YKBookStore",
79         "USER": "postgres",
80         "PASSWORD": "CP$3320$ummer!", # replace with your actual password
81         "HOST": "localhost",
82         "PORT": "5432",
83     }
84 }

```

6.2. Create Django Models for Your Tables

Replace the existing DATABASES configuration with the following:

Create a new file models.py



#models.py

from django.db import models

class Author(models.Model):

name = models.CharField(max_length=255)

birthdate = models.DateField()

nationality = models.CharField(max_length=255)

```
def __str__(self):  
    return self.name
```

```
class Book(models.Model):
```

```
    title = models.CharField(max_length=255)  
    author = models.ForeignKey(Author, on_delete=models.CASCADE)  
    genre = models.CharField(max_length=255)  
    price = models.DecimalField(max_digits=5, decimal_places=2)  
    publisher = models.CharField(max_length=255)  
    isbn = models.CharField(max_length=13)
```

```
def __str__(self):  
    return self.title
```

```
class Video(models.Model):
```

```
    title = models.CharField(max_length=255)  
    genre = models.CharField(max_length=255)  
    price = models.DecimalField(max_digits=5, decimal_places=2)  
    publisher = models.CharField(max_length=255)  
    release_date = models.DateField()
```

```
def __str__(self):  
    return self.title
```

```
class Student(models.Model):
```

```
    name = models.CharField(max_length=255)  
    email = models.EmailField(unique=True)  
    birthdate = models.DateField()
```

```
def __str__(self):  
    return self.name
```

```
class Transaction(models.Model):
```

```
    student = models.ForeignKey(Student, on_delete=models.CASCADE)  
    book = models.ForeignKey(Book, null=True, blank=True, on_delete=models.SET_NULL)  
    video = models.ForeignKey(Video, null=True, blank=True, on_delete=models.SET_NULL)  
    borrowed_date = models.DateField(auto_now_add=True)  
    due_date = models.DateField()
```

```
def __str__(self):  
    return f"{self.student.name} borrowed {self.book.title if self.book else self.video.title}"
```

6.2. Create a superuser for accessing the Django admin interface.

Make and Apply Migrations:

python manage.py makemigrations

python manage.py migrate

```
PS C:\Users\galae\Desktop\00472-337513824 GS Backup 12.20.22 JC\KEAN_Lectures\3320\django\bookstore> python manage.py makemigrations
No changes detected
PS C:\Users\galae\Desktop\00472-337513824 GS Backup 12.20.22 JC\KEAN_Lectures\3320\django\bookstore> python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, contenttypes, sessions
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying sessions.0001_initial... OK
```

6.3. Create a superuser for accessing the Django admin interface.

Create a superuser to access the Django admin interface:

python manage.py createsuperuser

```
Applying sessions.0001_initial... OK
PS C:\Users\galae\Desktop\00472-337513824 GS Backup 12.20.22 JC\KEAN_Lectures\3320\django\bookstore> python manage.py createsuperuser
Username (leave blank to use 'galae'): ykumar
Email address: ykumar@kean.edu
Password:
```

```
This password is too common.
Bypass password validation and create user anyway? [y/N]: y
Superuser created successfully.
```

Reference: <https://www.geeksforgeeks.org/how-to-create-superuser-in-django/>

7. Define New Models

7.1. Placeholder for a table

Come up with your own idea what tables can be present and belong to this database

Below is just a placeholder for you.

```
class NewModel(models.Model):
    field1 = models.CharField(max_length=255)
    field2 = models.IntegerField()
    # Add more fields as necessary

    def __str__(self):
        return self.field1
```

7.2. Run migrations

`python manage.py makemigrations`

`python manage.py migrate`

```
PS C:\Users\galae\Desktop\00472-337513824 GS Backup 12.20.22 JC\KEAN_Lectures\3320\django\bookstore> python manage.py makemigrations
No changes detected
PS C:\Users\galae\Desktop\00472-337513824 GS Backup 12.20.22 JC\KEAN_Lectures\3320\django\bookstore> python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, contenttypes, sessions
Running migrations:
  No migrations to apply.
```

Task 5. Screenshot your migrations code and migrations themselves and the output. They will appear in your project after running the command `python manage.py makemigrations`

and disappear after `python manage.py migrate`

8. Create Views and URLs with ChatGPT

Work with ChatGPT (or another LLM of your choice) to create views, template .html files and URLs for your project – This is a GUI of your app. Recall your work on the Python game and use the same approach. The goal is to display the database table content on the screen and be able to add, edit, display and delete data from the screen / GUI (not on the database side), at the same time you should see the changes on the database side as well.

To make your models accessible in the Django admin interface, register them in **admin.py**:

```
from django.contrib import admin
from .models import NewModel
```

```
admin.site.register(NewModel)
```

```
django > bookstore > bookstore > admin.py
1  from django.contrib import admin
2  from .models import NewModel
3
4  admin.site.register(NewModel)
```

7. Define New Models

Create views to handle the data and map them to URLs. In **views.py**:

```
from django.shortcuts import render
from .models import NewModel
```

```
def new_model_list(request):
    items = NewModel.objects.all()
    return render(request, 'new_model_list.html', {'items': items})
```

In **urls.py**:

```
from django.urls import path
from . import views
```

```
urlpatterns = [
    path('new_model/', views.new_model_list, name='new_model_list'),
]
```

Under `templates/new_model_list.html`:

```
<h1>New Model List</h1>
<ul>
    {% for item in items %}
        <li>{{ item.field1 }} - {{ item.field2 }}</li>
    {% endfor %}
</ul>
```

7. Views, Templates and URLs

Into views:

```
django > bookstore > bookstore > views.py > ...
1  from django.shortcuts import render
2  from .models import NewModel, Author, Book, Video
3
4  def new_model_list(request):
5      items = NewModel.objects.all()
6      return render(request, 'new_model_list.html', {'items': items})
7
8  def index(request):
9      books = Book.objects.all()
10     authors = Author.objects.all()
11     videos = Video.objects.all()
12     return render(request, 'bookstore/index.html', {'books': books, 'authors': authors, 'videos': videos})
```

Into urls:

```

django > bookstore > bookstore >  urls.py > ...
1  from django.contrib import admin
2  from django.urls import path
3  from . import views
4
5  urlpatterns = [
6      path("admin/", admin.site.urls),
7  ]
8
9  urlpatterns = [
10     path('new_model/', views.new_model_list, name='new_model_list'),
11 ]
12
13 urlpatterns = [
14     path('', views.index, name='index'),
15 ]

```

Html example:

Index.html:

```

<!DOCTYPE html>
<html>
<head>
    <title>Bookstore</title>
</head>
<body>
    <h1>Index Page</h1>

    <h1>Authors</h1>
    <ul>
        {% for author in authors %}
        <li>{{ author.name }}</li>
        {% endfor %}
    </ul>
</body>
</html>

```

new_model.html

```

<!DOCTYPE html>
<html>
<head>
    <title>New Models</title>
</head>
<body>
    <h1>New Models</h1>
    <ul>
        {% for new_model in new_models %}
        <li>{{ new_model.title }} - {{ new_model.description }}</li>

```

```
        {% endfor %}
    </ul>
</body>
</html>
```

Task 6. Take screenshot of your web page(s), they should display the exact data from your database. Screenshot your data in the database next to it to confirm this was not hardcoded.

8. Run the server

`python manage.py runserver`

Follow the link

```
PS C:\Users\galae\Desktop\00472-337513824 GS Backup 12.20.22 JC\KEAN_Lectures\3320\django\bookstore> python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).
June 15, 2024 - 20:27:38
Django version 5.0.6, using settings 'bookstore.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.
```

Task 7. Come up with your own idea what additional table(s) can be present in this project and belong to this database, add at least one table, model and view to this project related that was not there before.

Submit your work as a single PDF and a .zip file with the code